

Considerations for categorical data

EXPLORATORY DATA ANALYSIS IN PYTHON



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Why perform EDA?

- Detecting patterns and relationships
- Generating questions, or **hypotheses**
- Preparing data for machine learning



¹ Image credit: <https://unsplash.com/@simonesecci>

Representative data

- Sample represents the population

For example:

- Education versus income in USA
 - Can't use data from France

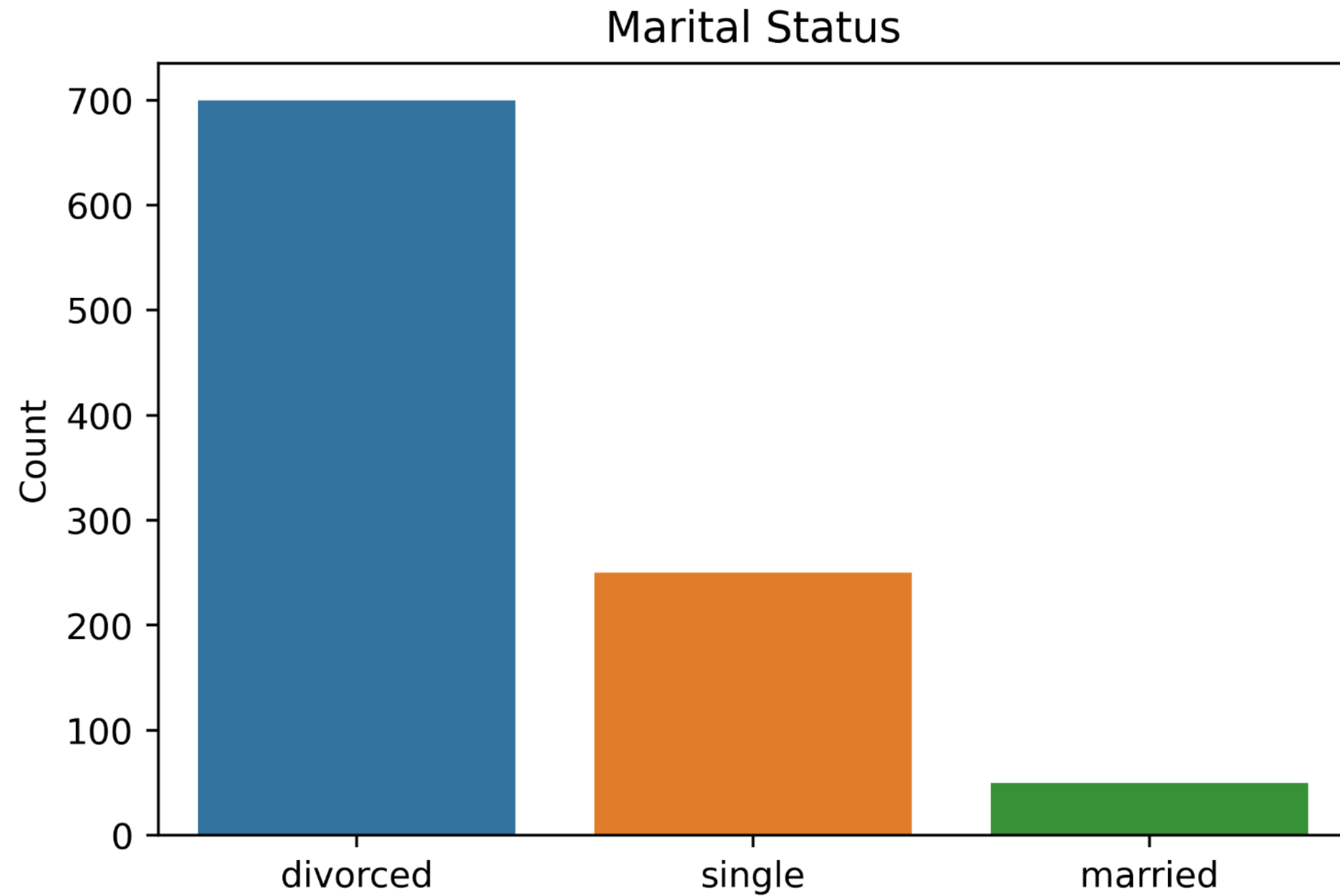


¹ Image credits: https://unsplash.com/@cristina_glebova; https://unsplash.com/@nimbus_vulpis

Categorical classes

- Classes = labels
- Survey people's attitudes towards marriage
 - Marital status
 - Single
 - Married
 - Divorced

Class imbalance



Class frequency

```
print(planes["Destination"].value_counts())
```

```
Cochin      4391
Bangalore   2773
Delhi        1219
New Delhi    888
Hyderabad    673
Kolkata      369
Name: Destination, dtype: int64
```

Relative class frequency

- 40% of internal Indian flights have a destination of Delhi

```
planes["Destination"].value_counts(normalize=True)
```

```
Cochin      0.425773
Bangalore   0.268884
Delhi        0.118200
New Delhi   0.086105
Hyderabad    0.065257
Kolkata      0.035780
Name: Destination, dtype: float64
```

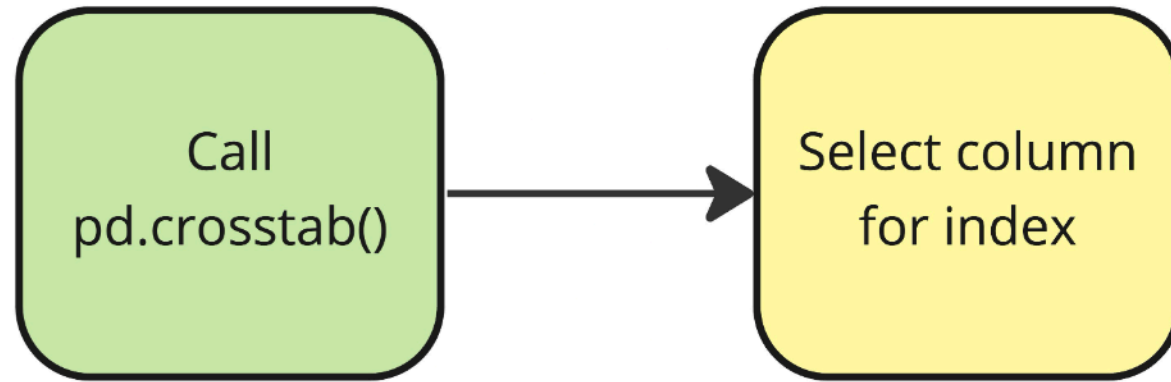
- Is our sample representative of the population (Indian internal flights)?

Cross-tabulation

Call
`pd.crosstab()`

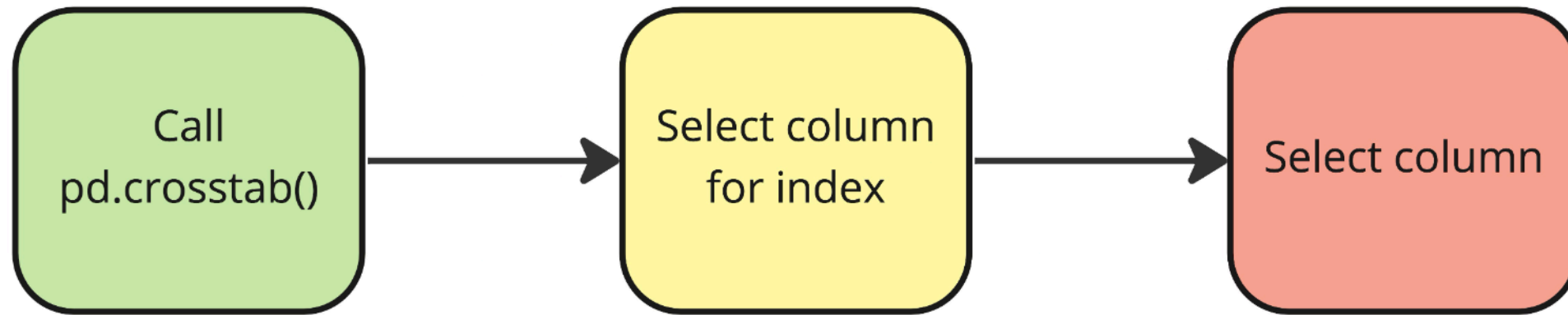
```
pd.crosstab(
```

Select index



```
pd.crosstab(planes["Source"],
```

Select columns



```
pd.crosstab(planes["Source"], planes["Destination"])
```

Cross-tabulation

Destination Source	Banglore	Cochin	Delhi	Hyderabad	Kolkata	New Delhi
Banglore	0	0	1199	0	0	868
Chennai	0	0	0	0	364	0
Delhi	0	4318	0	0	0	0
Kolkata	2720	0	0	0	0	0
Mumbai	0	0	0	662	0	0

Extending cross-tabulation

Source	Destination	Median Price (IDR)
Banglore	Delhi	4232.21
Banglore	New Delhi	12114.56
Chennai	Kolkata	3859.76
Delhi	Cochin	9987.63
Kolkata	Banglore	9654.21
Mumbai	Hyderabad	3431.97

Aggregated values with pd.crosstab()

```
pd.crosstab(planes["Source"], planes["Destination"],  
            values=planes["Price"], aggfunc="median")
```

Destination	Banglore	Cochin	Delhi	Hyderabad	Kolkata	New Delhi
Source						
Banglore	NaN	NaN	4823.0	NaN	NaN	10976.5
Chennai	NaN	NaN	NaN	NaN	3850.0	NaN
Delhi	NaN	10262.0	NaN	NaN	NaN	NaN
Kolkata	9345.0	NaN	NaN	NaN	NaN	NaN
Mumbai	NaN	NaN	NaN	3342.0	NaN	NaN

Comparing sample to population

Source	Destination	Median Price (IDR)	Median Price (dataset)
Banglore	Delhi	4232.21	4823.0
Banglore	New Delhi	12114.56	10976.50
Chennai	Kolkata	3859.76	3850.0
Delhi	Cochin	9987.63	10260.0
Kolkata	Banglore	9654.21	9345.0
Mumbai	Hyderabad	3431.97	3342.0

Let's practice!

EXPLORATORY DATA ANALYSIS IN PYTHON

Generating new features

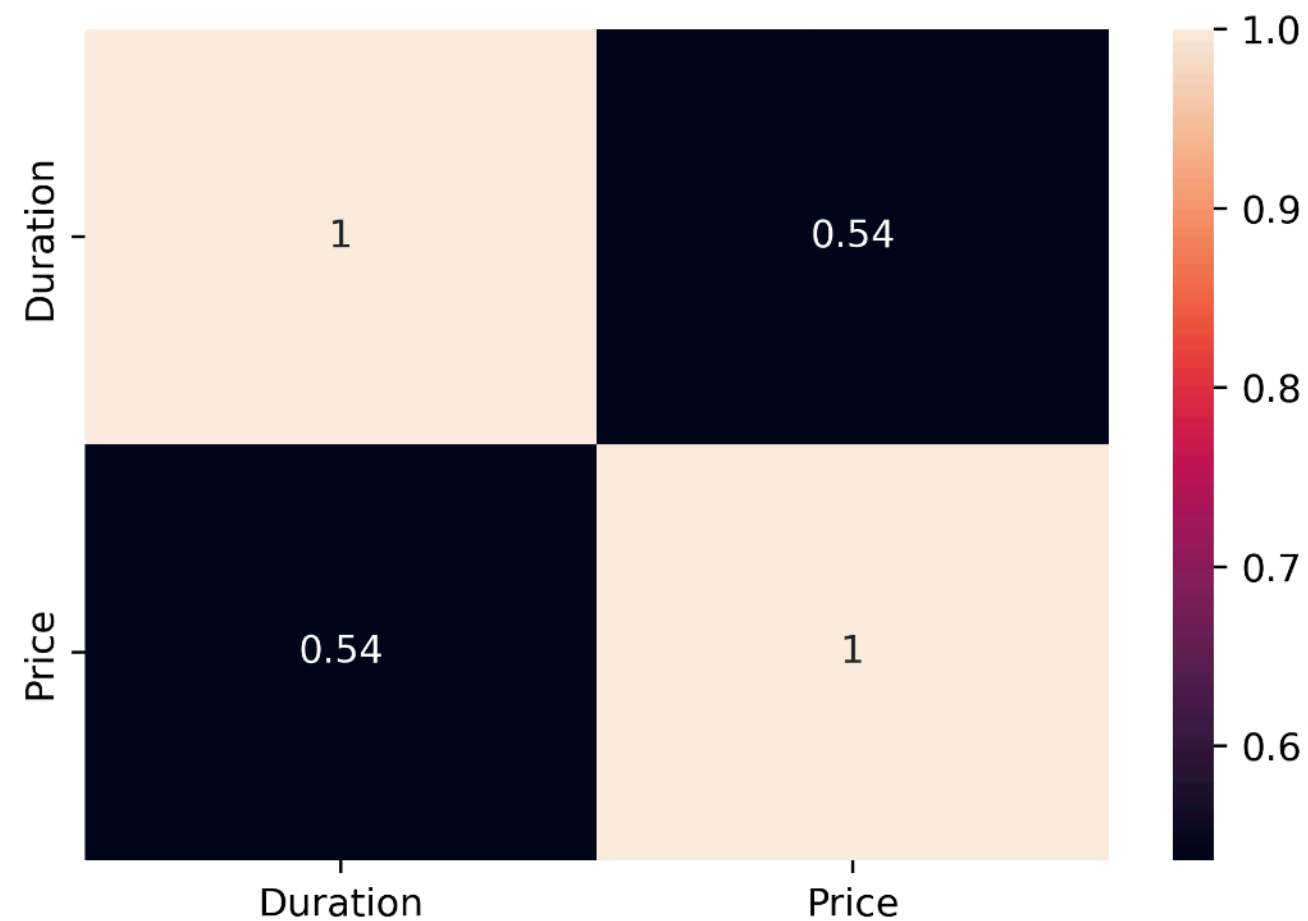
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Correlation

```
sns.heatmap(planes.corr(numeric_only=True), annot=True)  
plt.show()
```



Viewing data types

```
print(planes.dtypes)
```

```
Airline          object
Date_of_Journey  datetime64[ns]
Source           object
Destination      object
Route           object
Dep_Time         datetime64[ns]
Arrival_Time     datetime64[ns]
Duration         float64
Total_Stops      object
Additional_Info   object
Price           float64
dtype: object
```


Total stops

```
print(planes["Total_Stops"].value_counts())
```

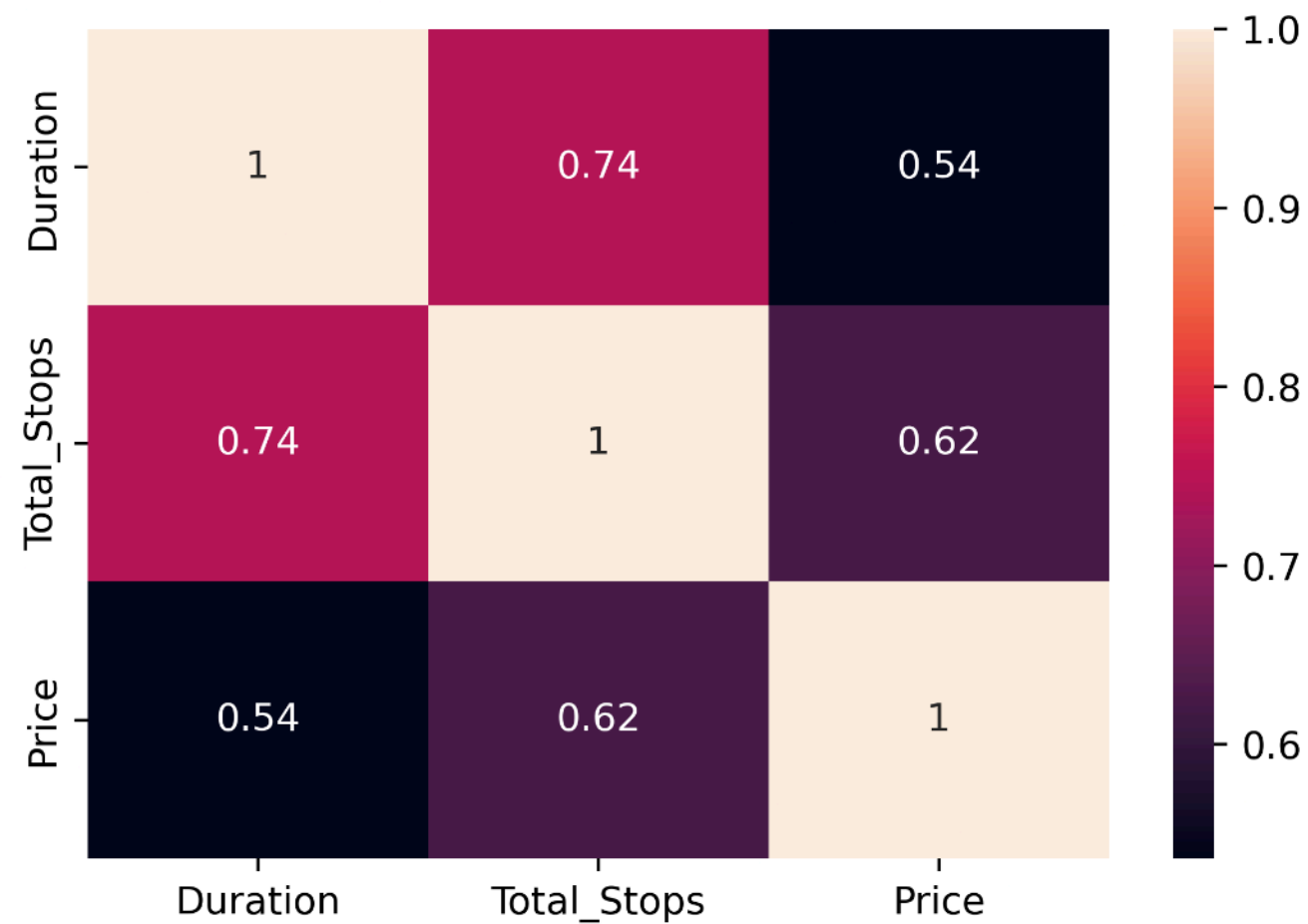
```
1 stop      4107
non-stop    2584
2 stops     1127
3 stops       29
4 stops        1
Name: Total_Stops, dtype: int64
```


Cleaning total stops

```
planes["Total_Stops"] = planes["Total_Stops"].str.replace(" stops", "")  
planes["Total_Stops"] = planes["Total_Stops"].str.replace(" stop", "")  
planes["Total_Stops"] = planes["Total_Stops"].str.replace("non-stop", "0")  
planes["Total_Stops"] = planes["Total_Stops"].astype(int)
```

Correlation

```
sns.heatmap(planes.corr(numeric_only=True), annot=True)  
plt.show()
```



Dates

```
print(planes.dtypes)
```

```
Airline          object
Date_of_Journey  datetime64[ns]
Source           object
Destination      object
Route            object
Dep_Time         datetime64[ns]
Arrival_Time     datetime64[ns]
Duration         float64
Total_Stops      int64
Additional_Info   object
Price            float64
dtype: object
```

Extracting month and weekday

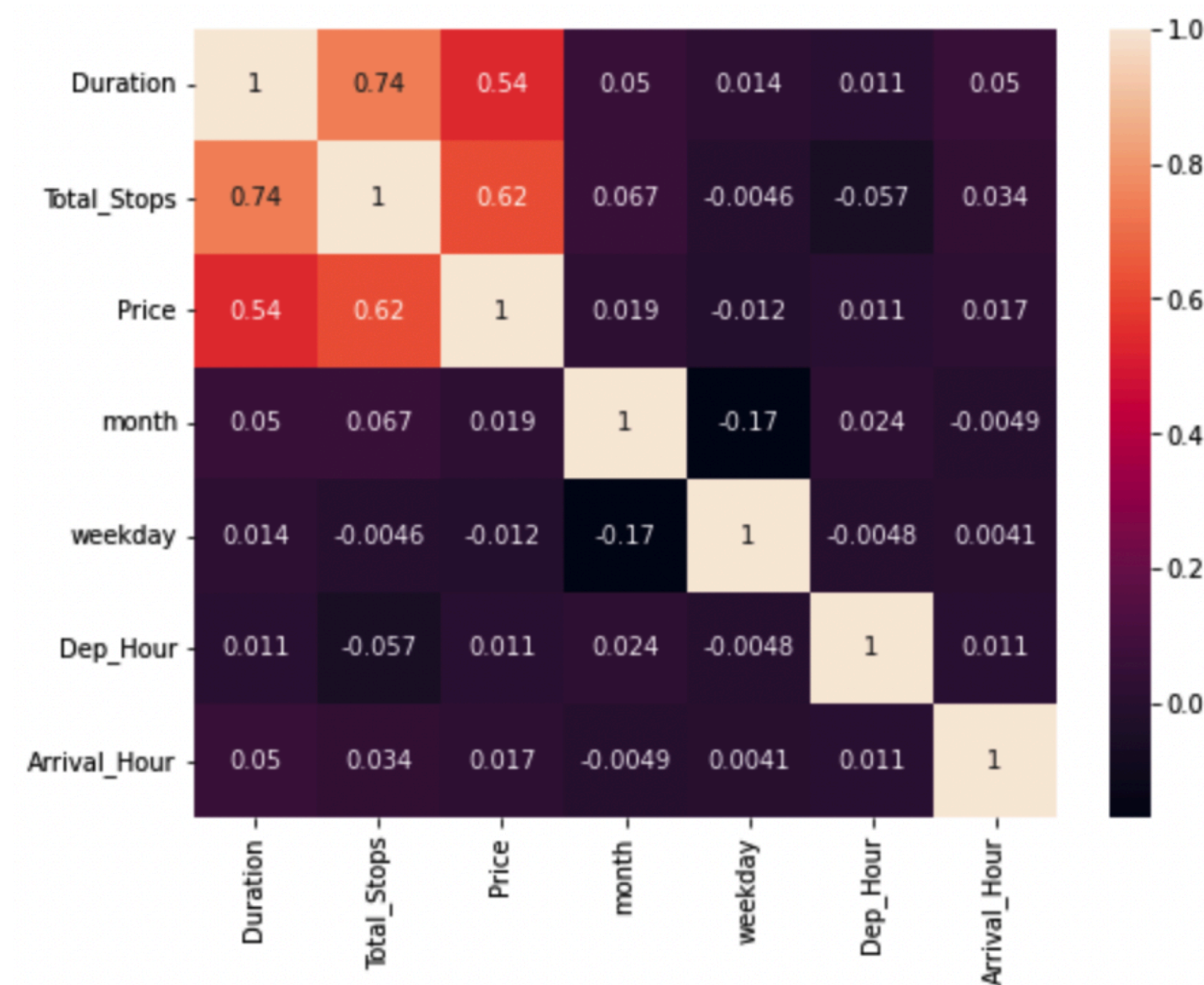
```
planes["month"] = planes["Date_of_Journey"].dt.month  
planes["weekday"] = planes["Date_of_Journey"].dt.weekday  
print(planes[["month", "weekday", "Date_of_Journey"]].head())
```

	month	weekday	Date_of_Journey
0	9	4	2019-09-06
1	12	3	2019-12-05
2	1	3	2019-01-03
3	6	0	2019-06-24
4	12	1	2019-12-03

Departure and arrival times

```
planes["Dep_Hour"] = planes["Dep_Time"].dt.hour  
planes["Arrival_Hour"] = planes["Arrival_Time"].dt.hour
```

Correlation



Creating categories

```
print(planes["Price"].describe())
```

```
count      7848.000000
mean       9035.413609
std        4429.822081
min        1759.000000
25%        5228.000000
50%        8355.000000
75%       12373.000000
max       54826.000000
Name: Price, dtype: float64
```

Range	Ticket Type
≤ 5228	Economy
$> 5228 \leq 8355$	Premium Economy
$> 8335 \leq 12373$	Business Class
> 12373	First Class

Descriptive statistics

```
twenty_fifth = planes["Price"].quantile(0.25)
median = planes["Price"].median()
seventy_fifth = planes["Price"].quantile(0.75)
maximum = planes["Price"].max()
```


Labels and bins

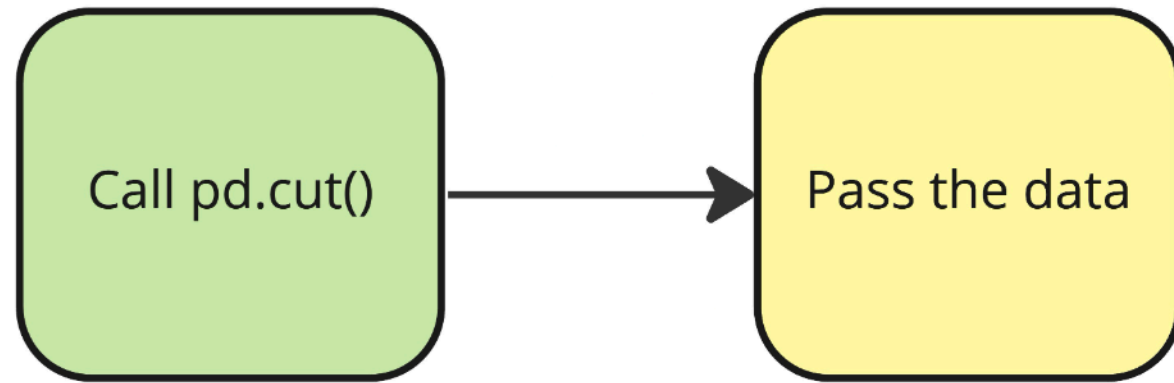
```
labels = ["Economy", "Premium Economy", "Business Class", "First Class"]  
bins = [0, twenty_fifth, median, seventy_fifth, maximum]
```

pd.cut()

Call pd.cut()

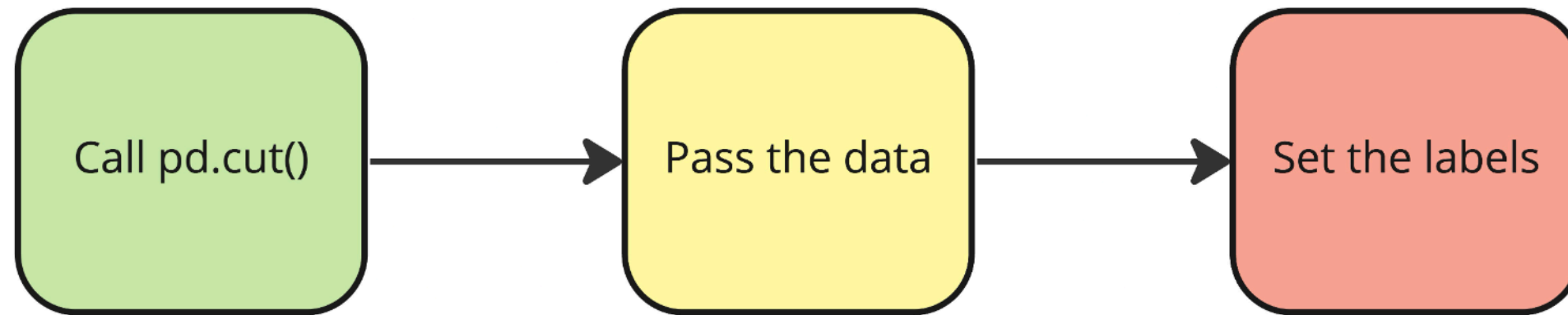
```
planes["Price_Category"] = pd.cut(
```

pd.cut()



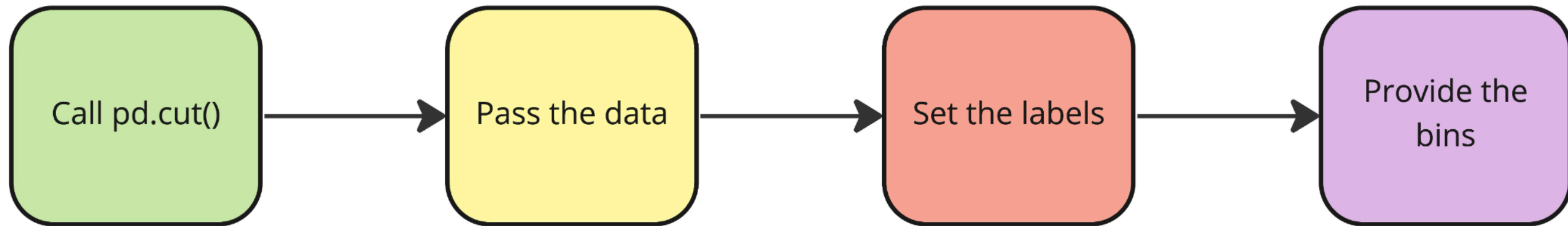
```
planes["Price_Category"] = pd.cut(planes["Price"],
```

pd.cut()



```
planes["Price_Category"] = pd.cut(planes["Price"],  
                                  labels=labels,
```

pd.cut()



```
planes["Price_Category"] = pd.cut(planes["Price"],  
                                  labels=labels,  
                                  bins=bins)
```

Price categories

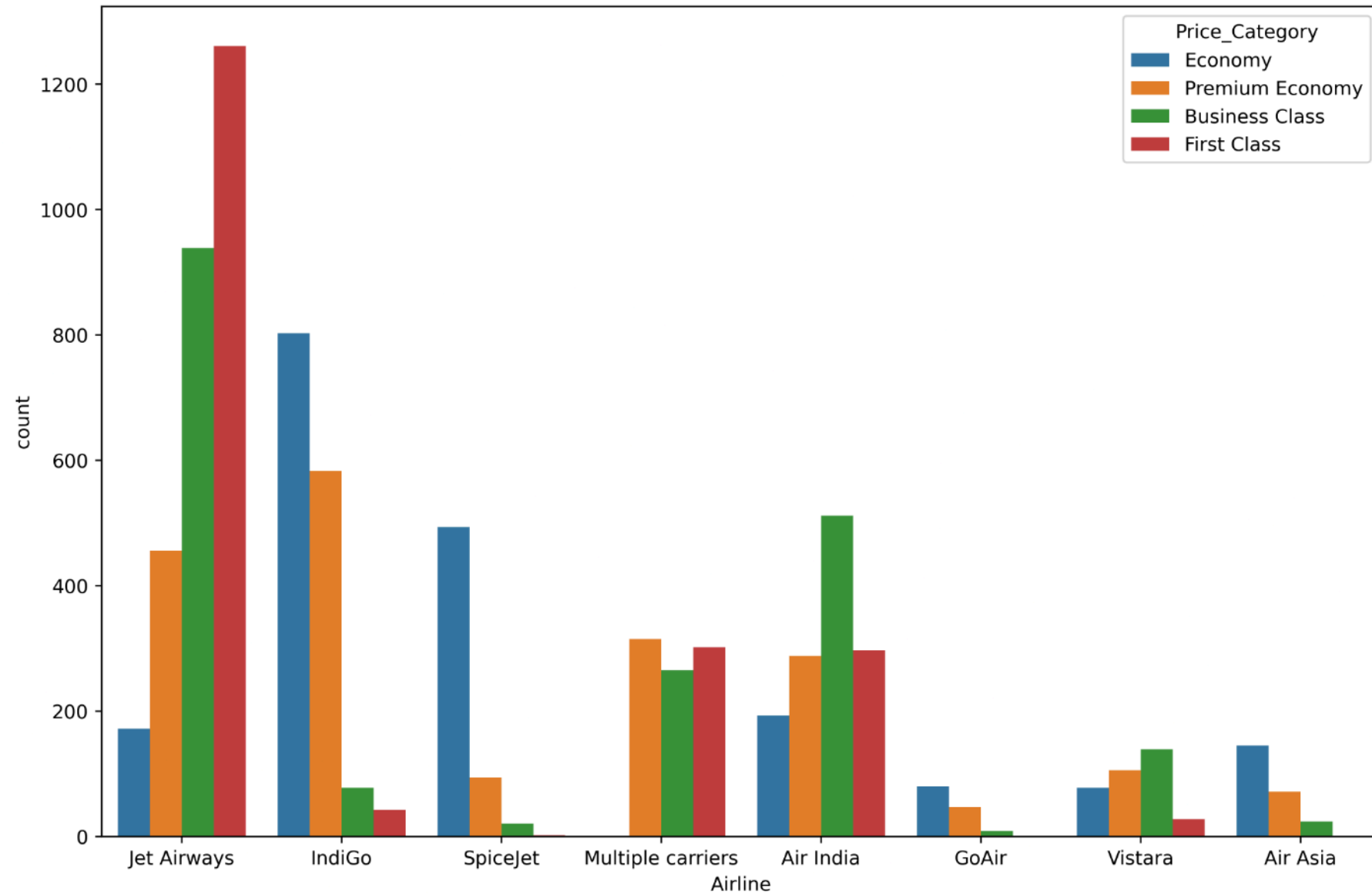
```
print(planes[["Price", "Price_Category"]].head())
```

	Price	Price_Category
0	13882.0	First Class
1	6218.0	Premium Economy
2	13302.0	First Class
3	3873.0	Economy
4	11087.0	Business Class

Price category by airline

```
sns.countplot(data=planes, x="Airline", hue="Price_Category")  
plt.show()
```


Price category by airline



Let's practice!

EXPLORATORY DATA ANALYSIS IN PYTHON

Generating hypotheses

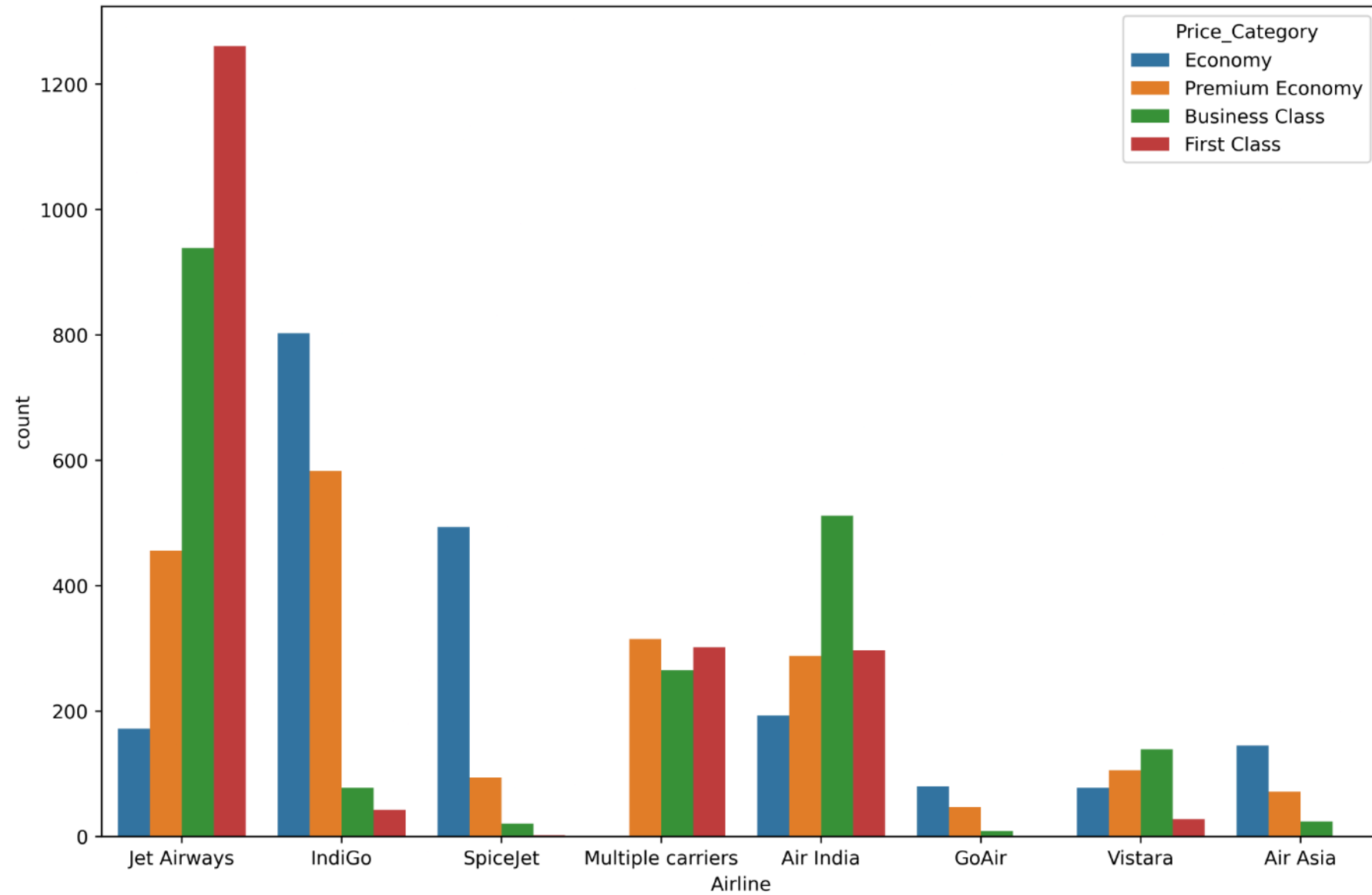
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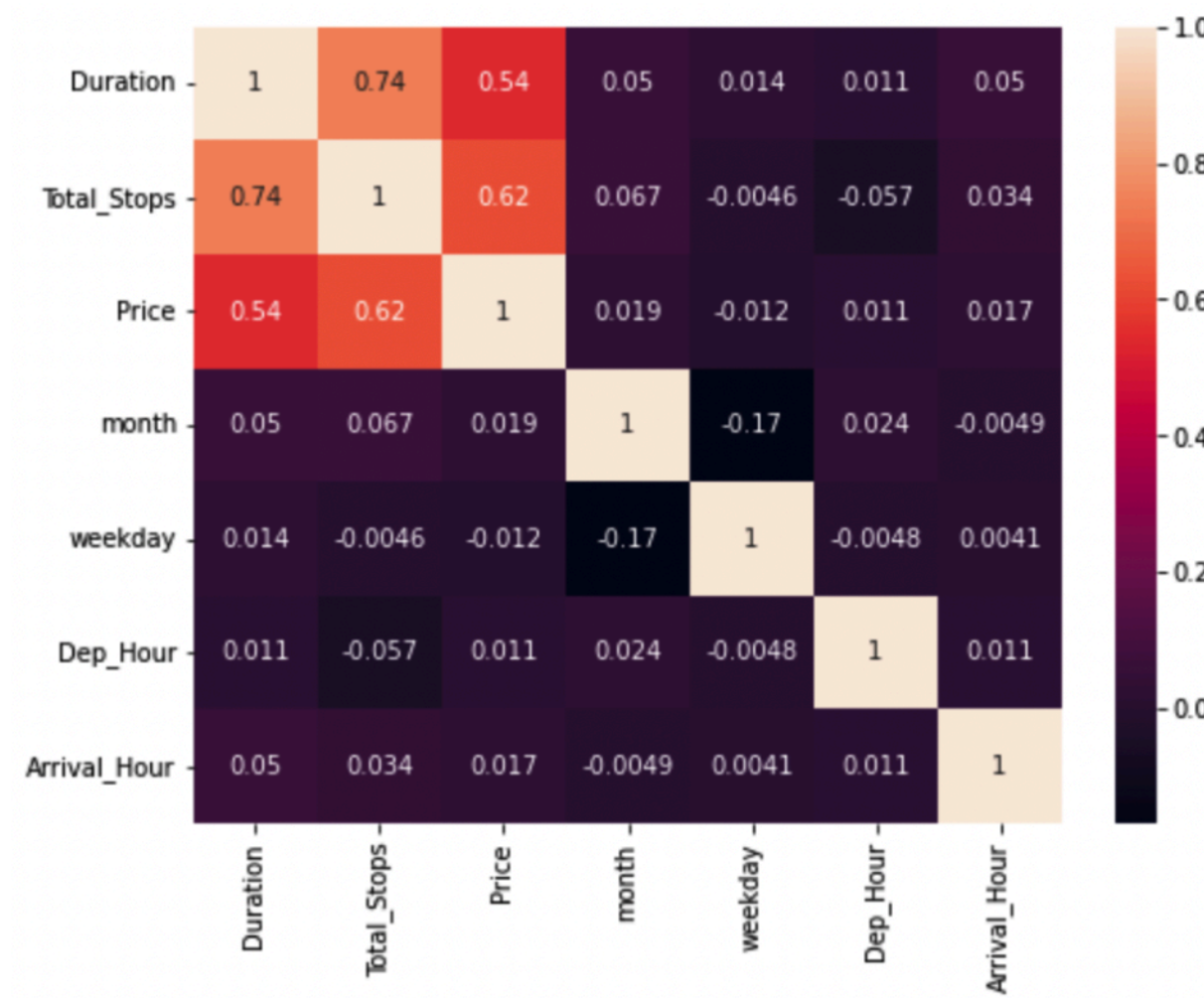
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What do we know?



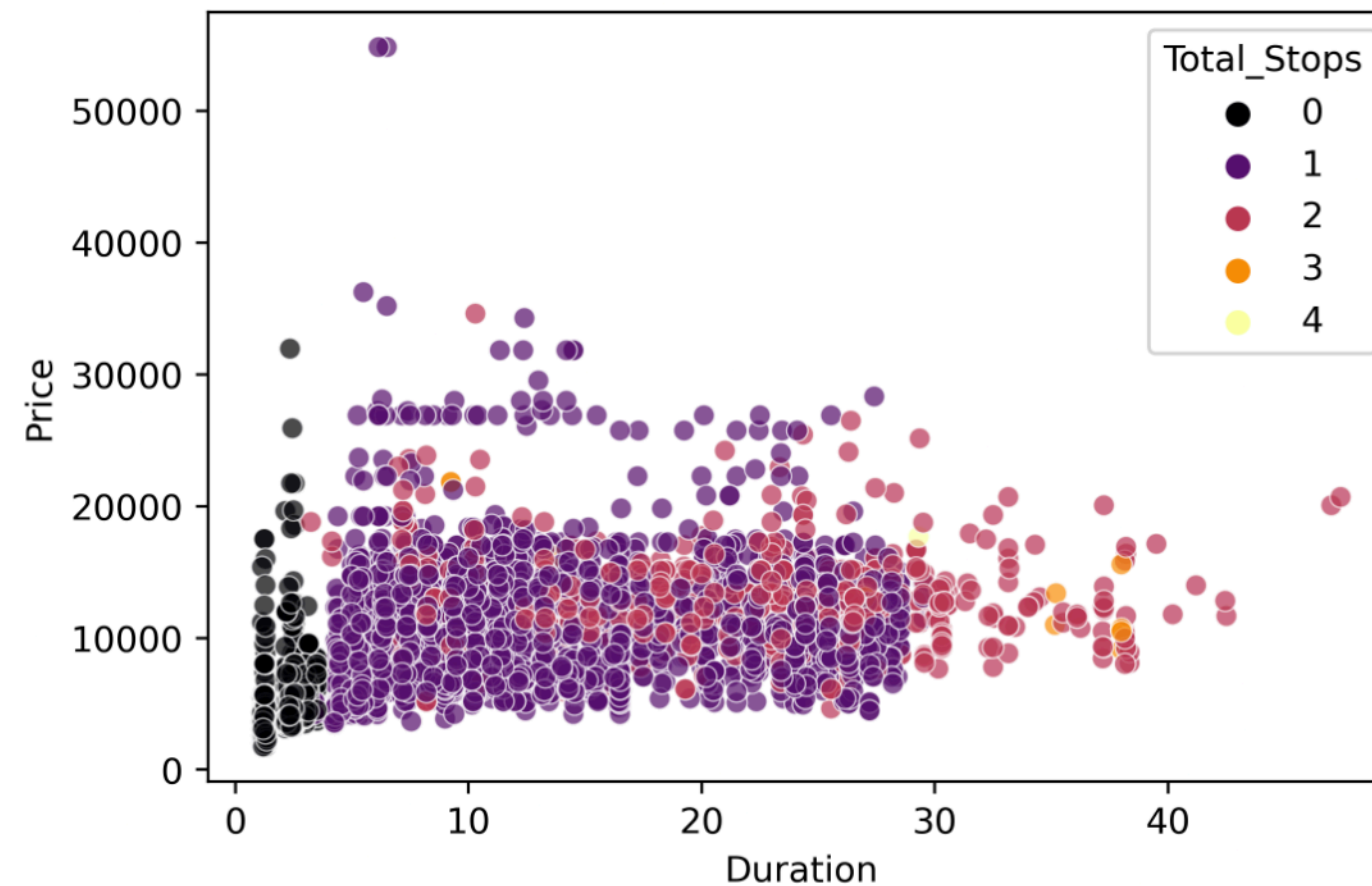
What do we know?

```
sns.heatmap(planes.corr(numeric_only=True), annot=True)  
plt.show()
```

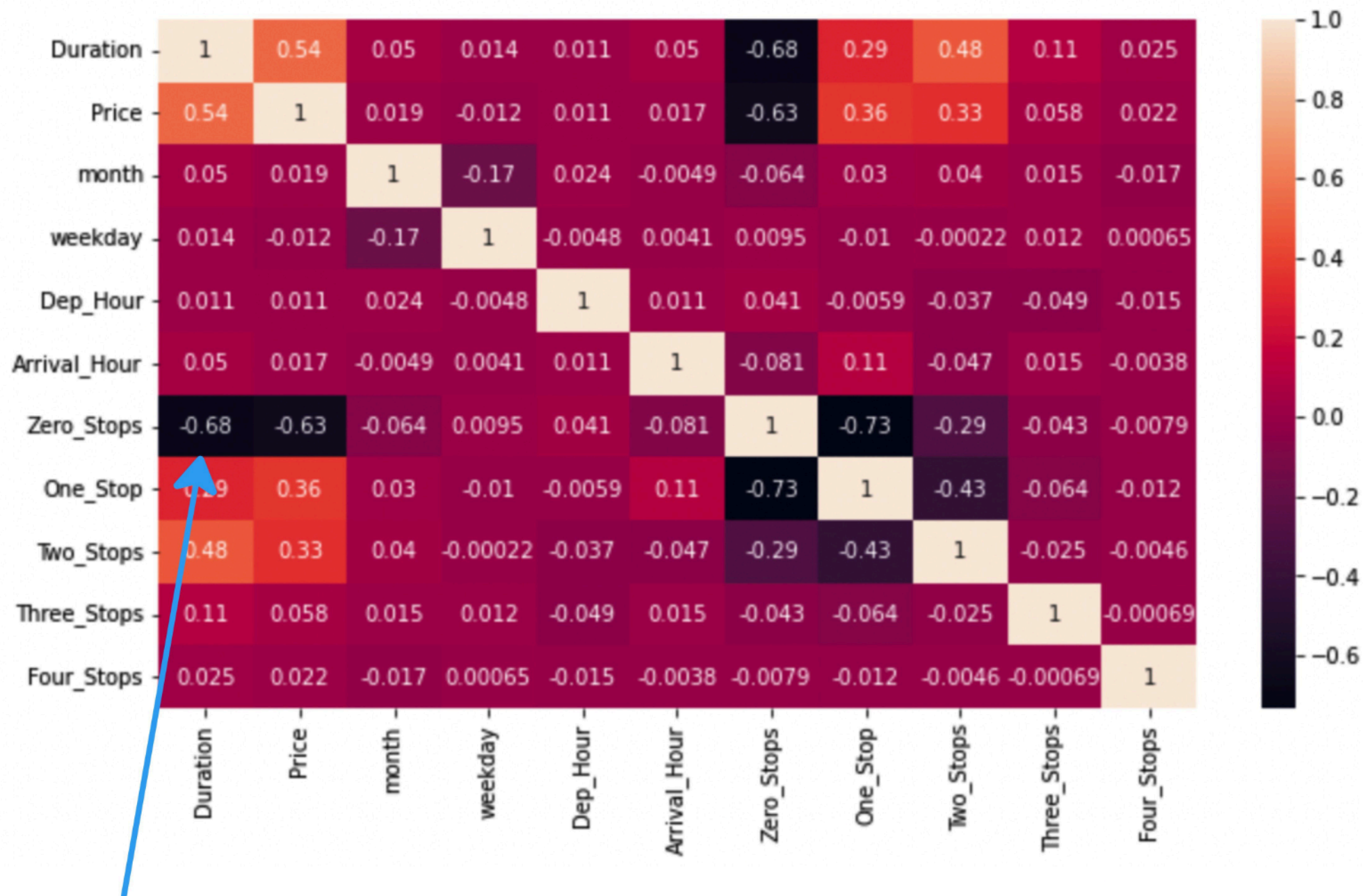


Spurious correlation

```
sns.scatterplot(data=planes, x="Duration", y="Price", hue="Total_Stops")  
plt.show()
```



How do we know?



What is true?



- Would data from a different time give the same results?
- Detecting relationships, differences, and patterns:
 - We use **Hypothesis Testing**
- Hypothesis testing requires, prior to data collection:
 - Generating a hypothesis or question
 - A decision on what statistical test to use

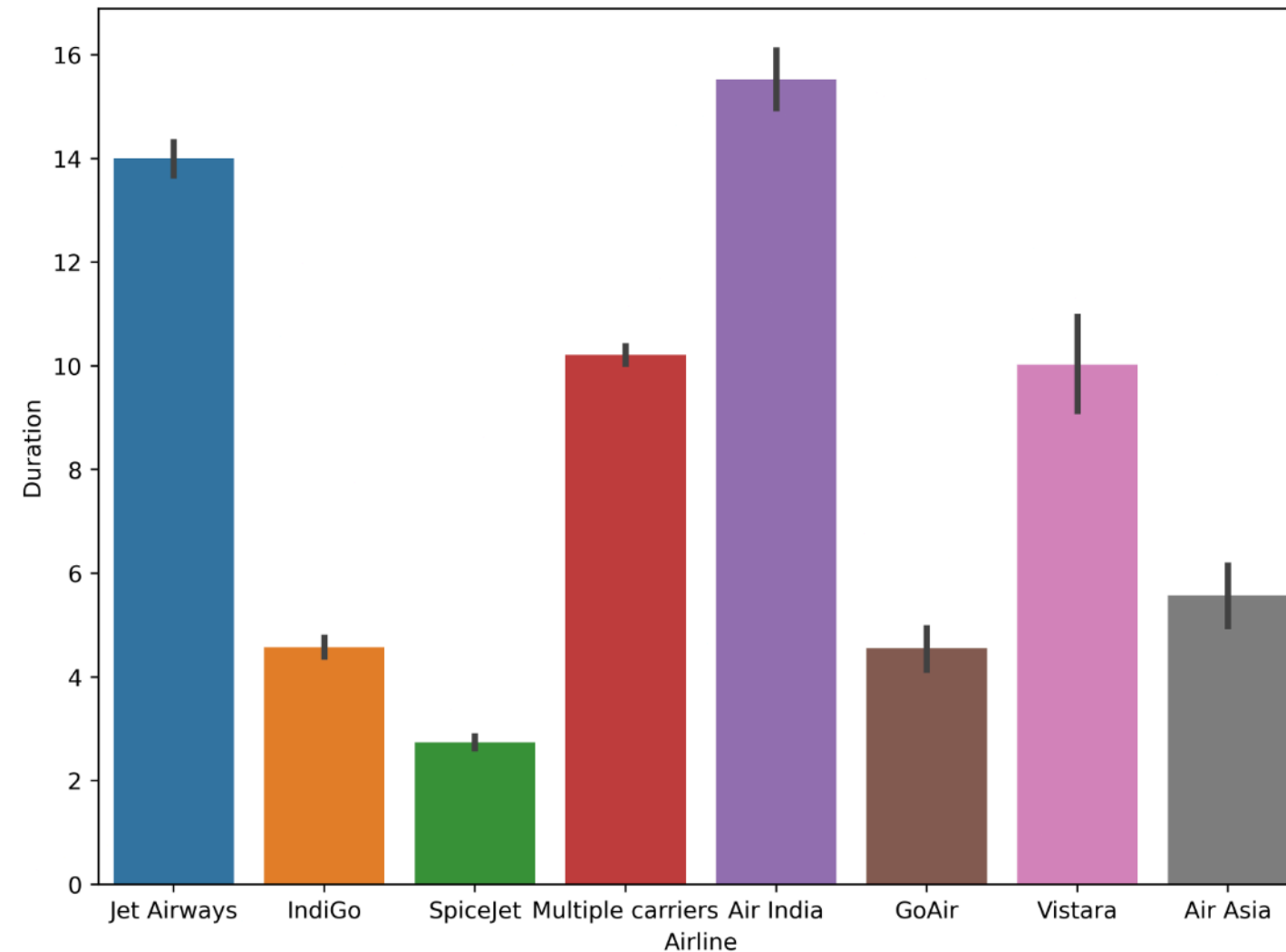
¹ Image credit: <https://unsplash.com/@markuswinkler>

Data snooping



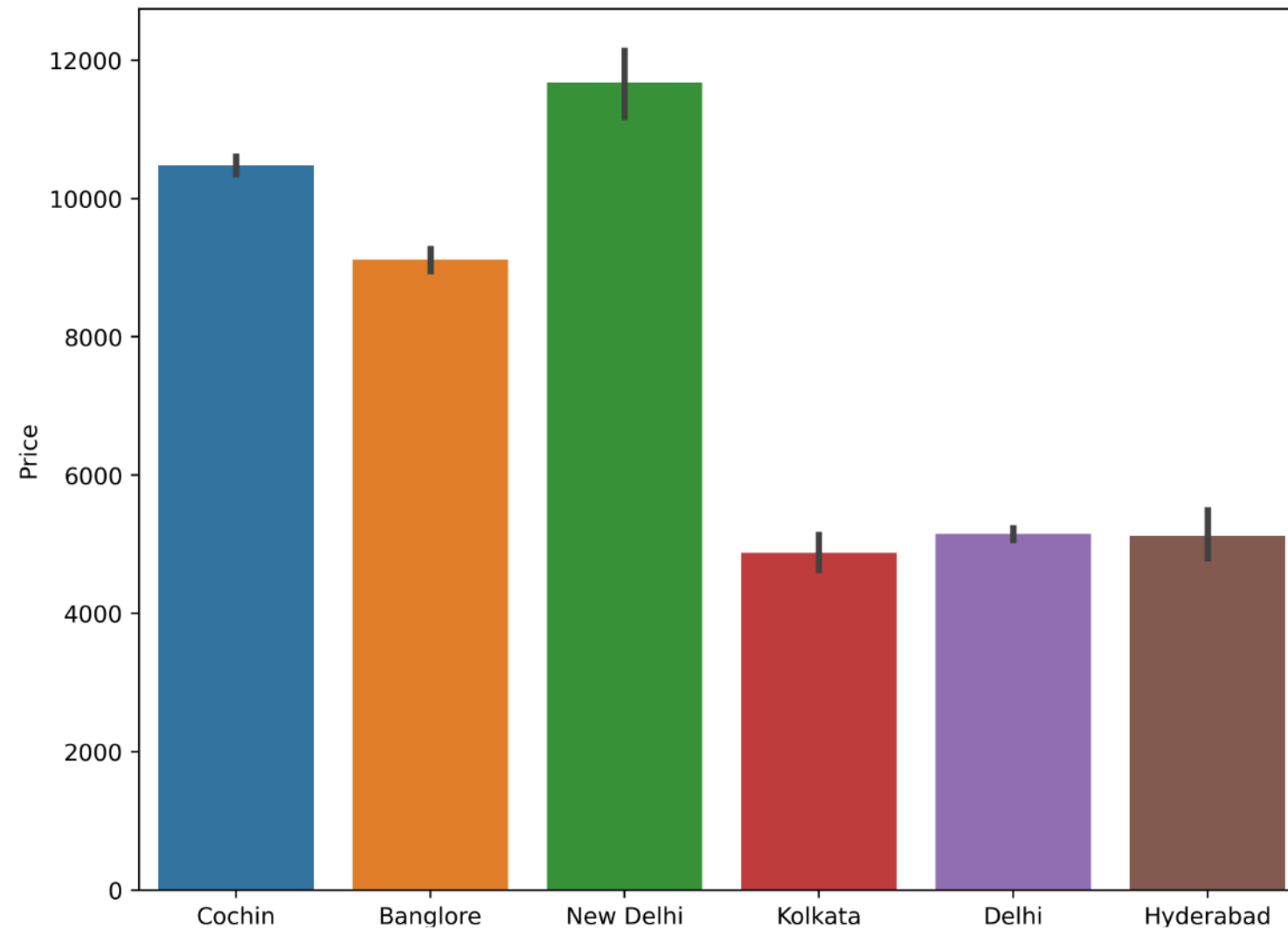
Generating hypotheses

```
sns.barplot(data=planes, x="Airline", y="Duration")  
plt.show()
```



Generating hypotheses

```
sns.barplot(data=planes, x="Destination", y="Price")  
plt.show()
```



Next steps

- Design our experiment
- Involves steps such as:
 - Choosing a sample
 - Calculating how many data points we need
 - Deciding what statistical test to run

Let's practice!

EXPLORATORY DATA ANALYSIS IN PYTHON

Congratulations

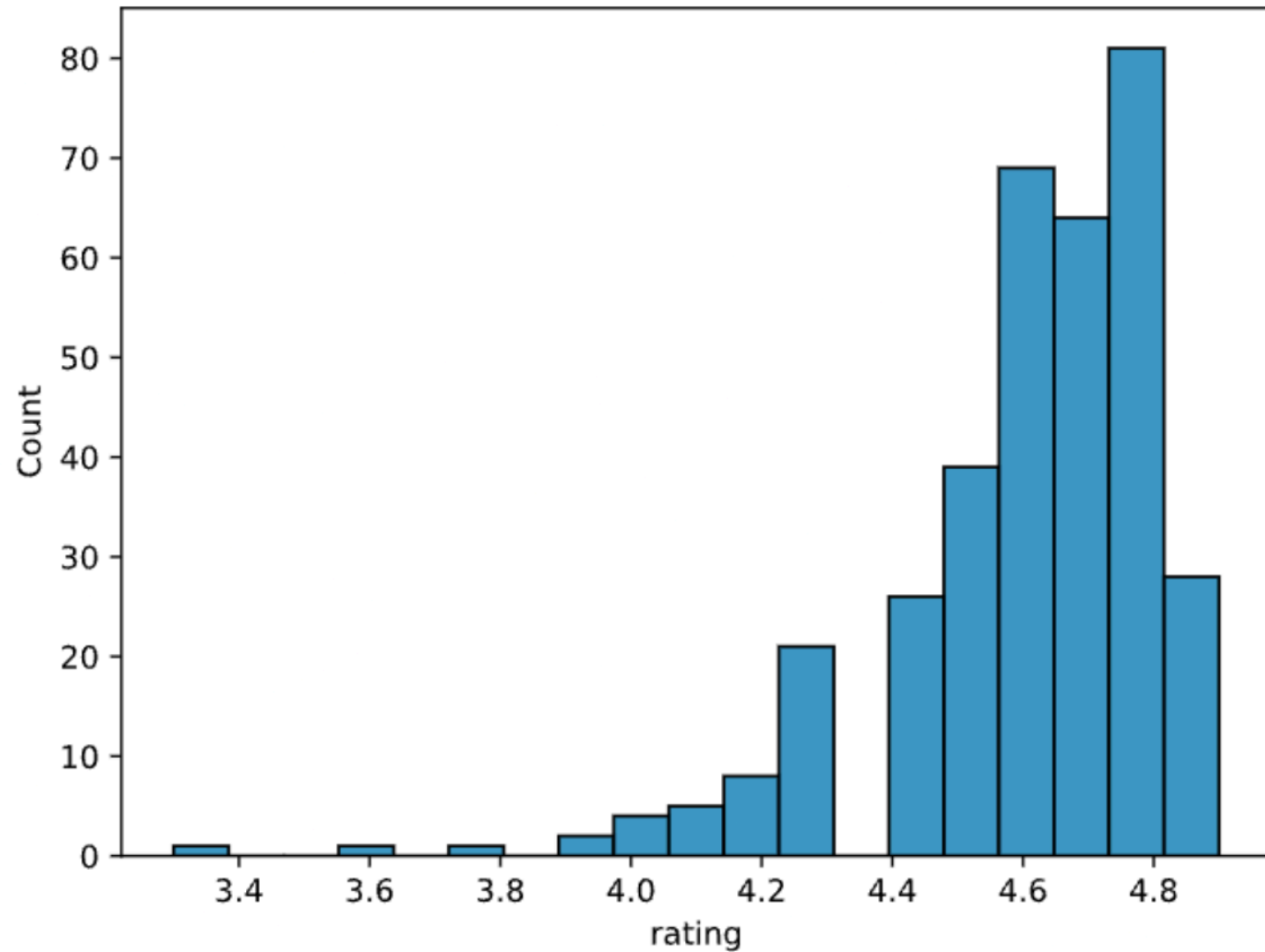
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Inspection and validation



```
books["year"] = books["year"].astype(int)
books.dtypes
```

```
name      object
author    object
rating     float64
year       int64
genre      object
dtype: object
```

Aggregation

```
books.groupby("genre").agg(  
    mean_rating=("rating", "mean"),  
    std_rating=("rating", "std"),  
    median_year=("year", "median")  
)
```

genre	mean_rating	std_rating	median_year
Childrens	4.780000	0.122370	2015.0
Fiction	4.570229	0.281123	2013.0
Non Fiction	4.598324	0.179411	2013.0

Address missing data

```
print(salaries.isna().sum())
```

```
Working_Year      12  
Designation       27  
Experience        33  
Employment_Status 31  
Employee_Location 28  
Company_Size      40  
Remote_Working_Ratio 24  
Salary_USD        60  
dtype: int64
```

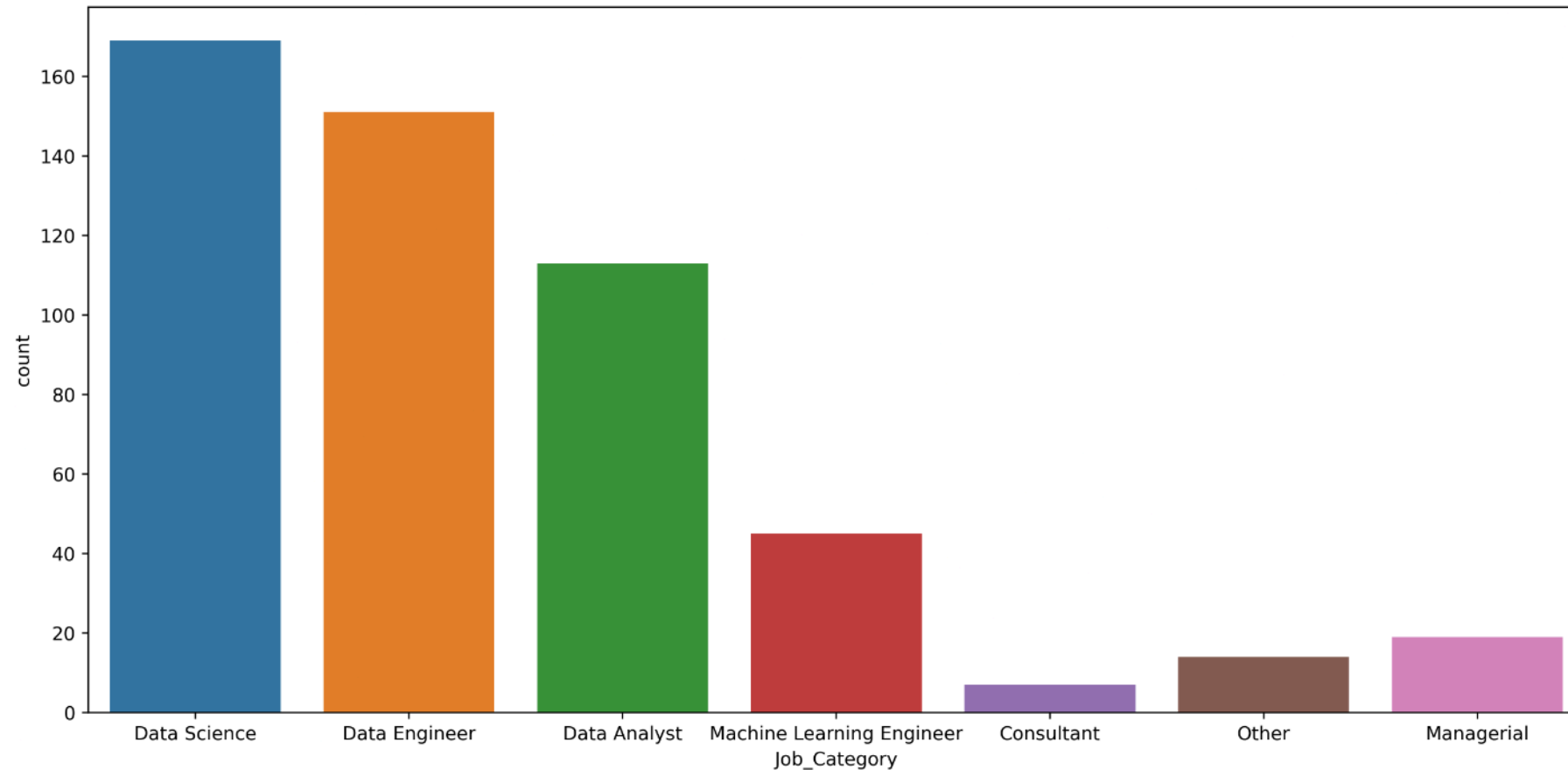

Address missing data

- Drop missing values
- Impute mean, median, mode
- Impute by sub-group

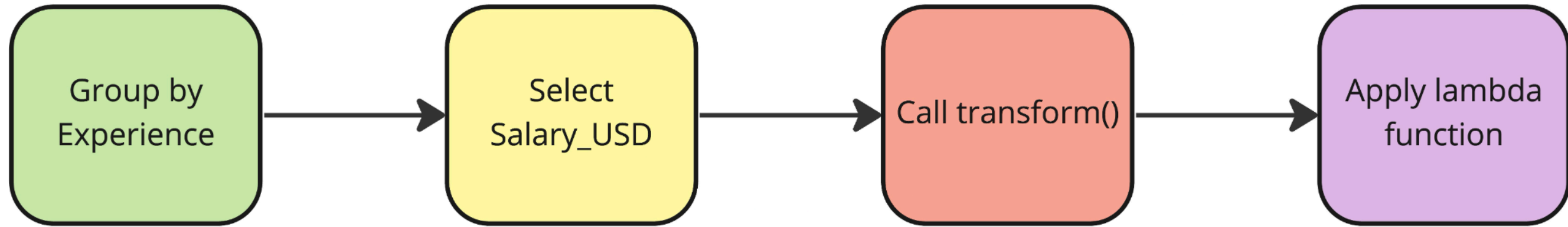
```
salaries_dict = salaries.groupby("Experience")["Salary_USD"].median().to_dict()
salaries["Salary_USD"] = salaries["Salary_USD"].fillna(salaries["Experience"].map(salaries_dict))
```

Analyze categorical data

```
salaries["Job_Category"] = np.select(conditions,  
                                     job_categories,  
                                     default="Other")
```



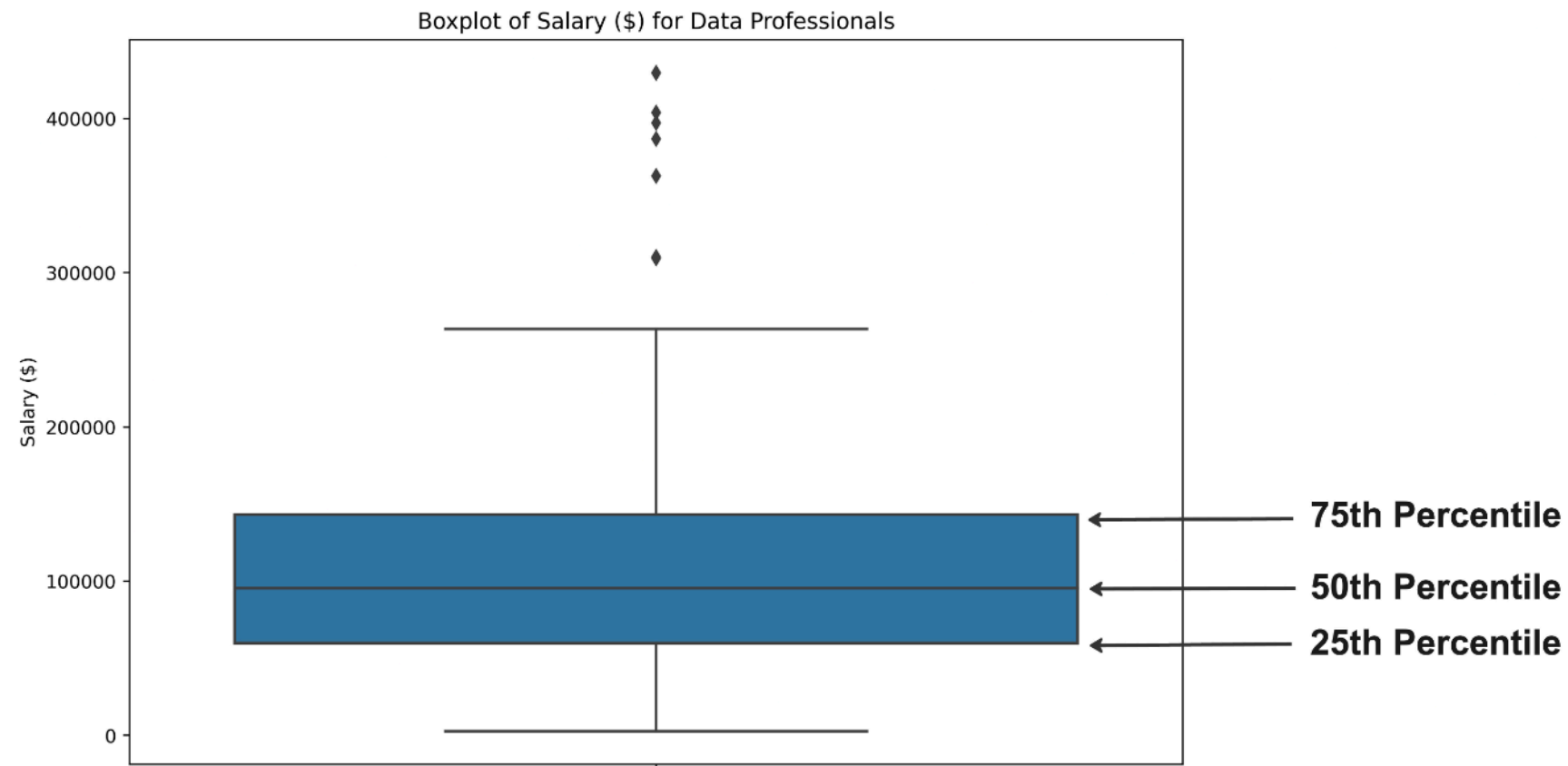
Apply lambda functions



```
salaries["std_dev"] = salaries.groupby("Experience")["Salary_USD"].transform(lambda x: x.std())
```

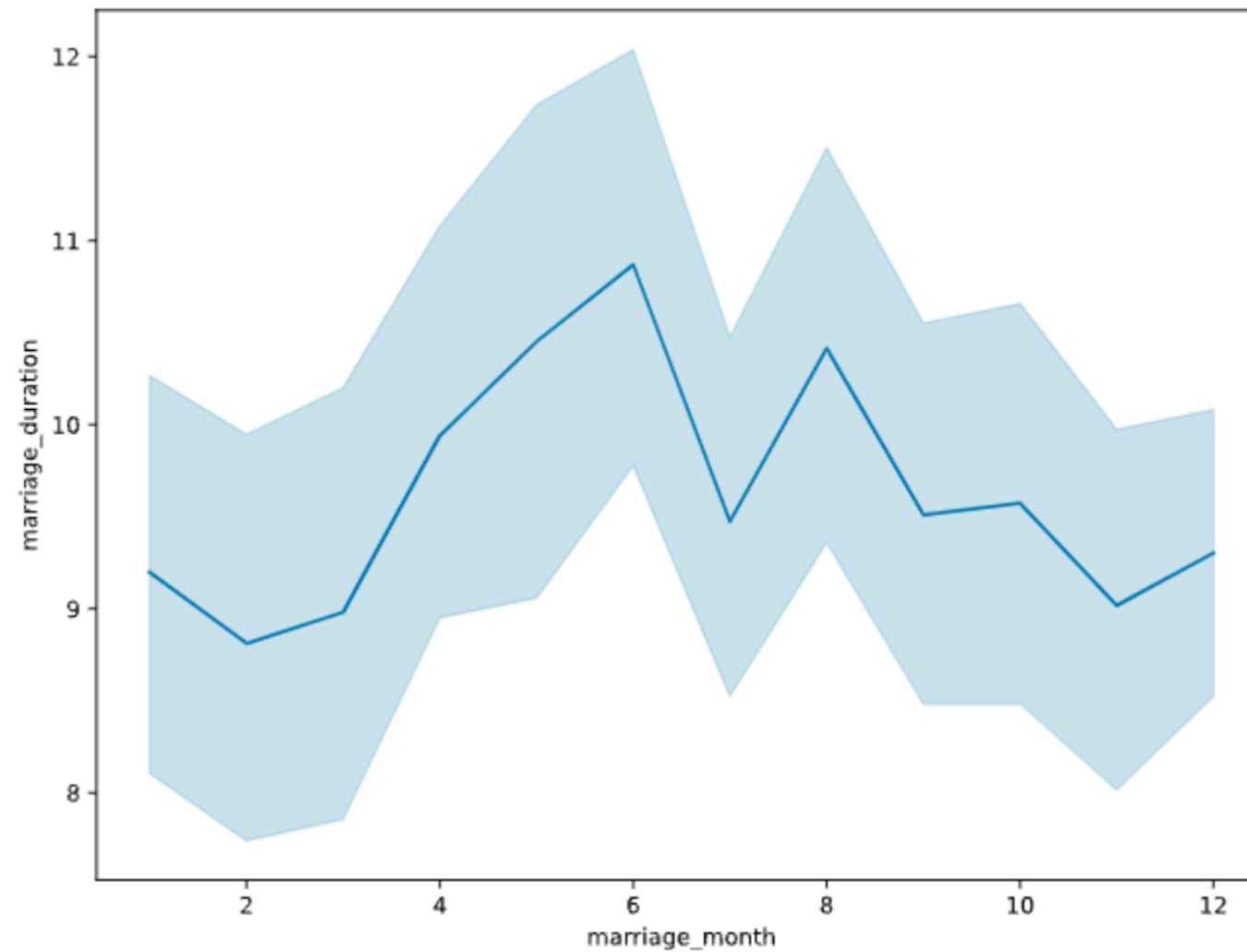
Handle outliers

```
sns.boxplot(data=salaries,  
            y="Salary_USD")  
  
plt.show()
```



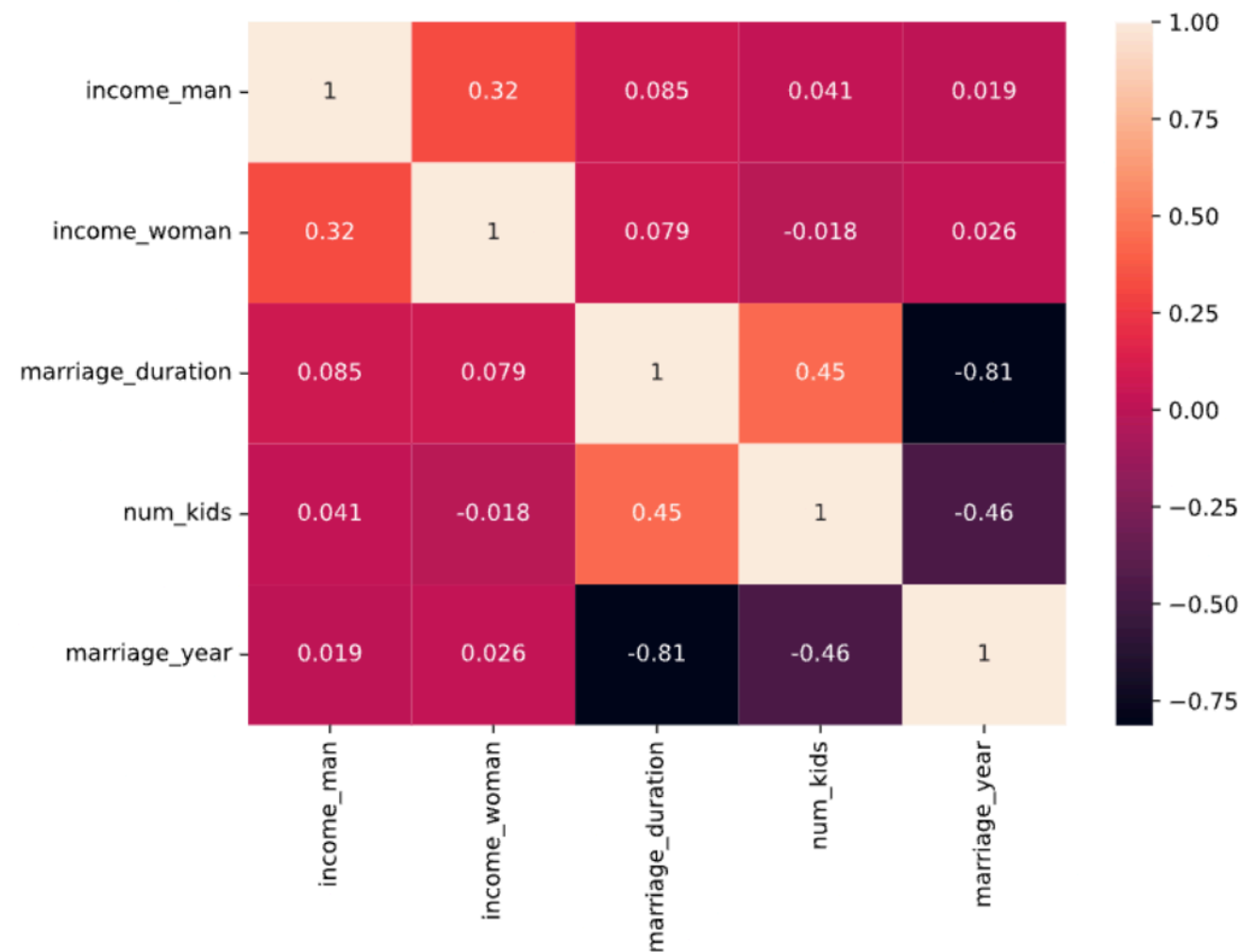
Patterns over time

```
sns.lineplot(data=divorce, x="marriage_month", y="marriage_duration")  
plt.show()
```



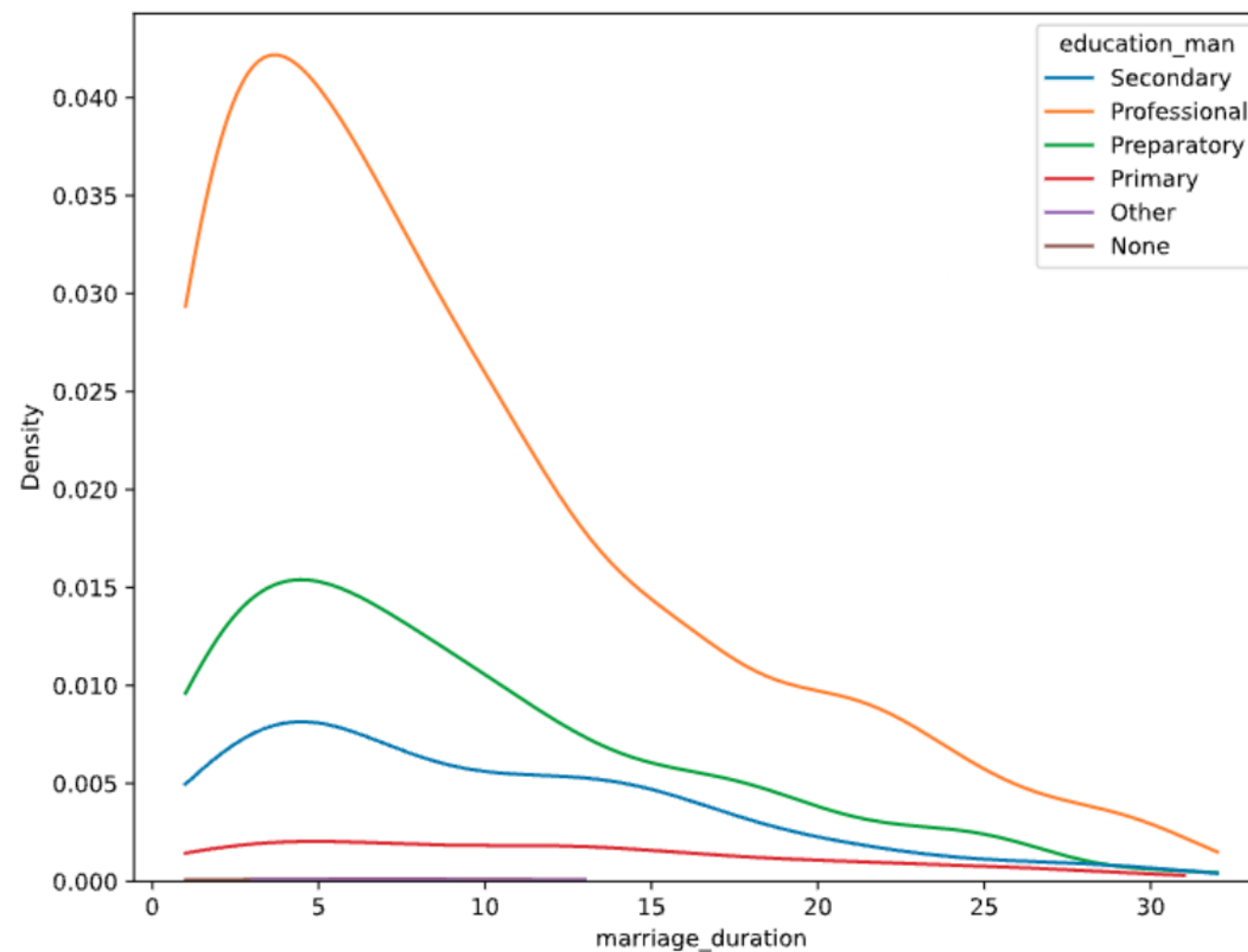
Correlation

```
sns.heatmap(divorce.corr(numeric_only=True), annot=True)  
plt.show()
```



Distributions

```
sns.kdeplot(data=divorce, x="marriage_duration", hue="education_man", cut=0)  
plt.show()
```

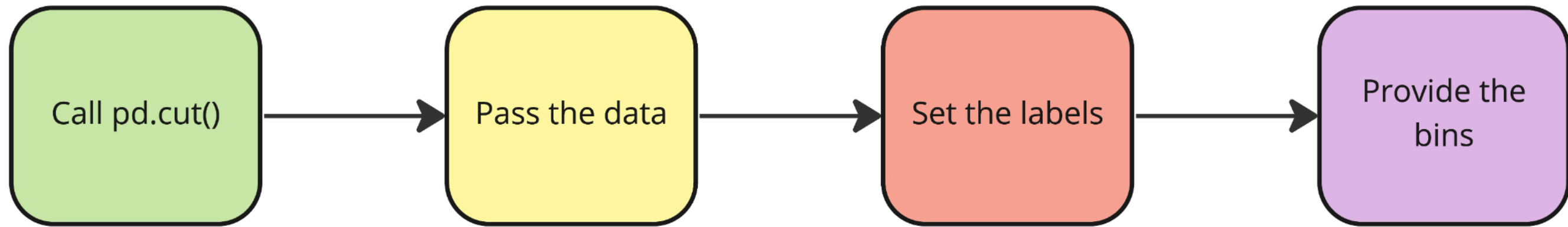


Cross-tabulation

```
pd.crosstab(planes["Source"], planes["Destination"],  
            values=planes["Price"], aggfunc="median")
```

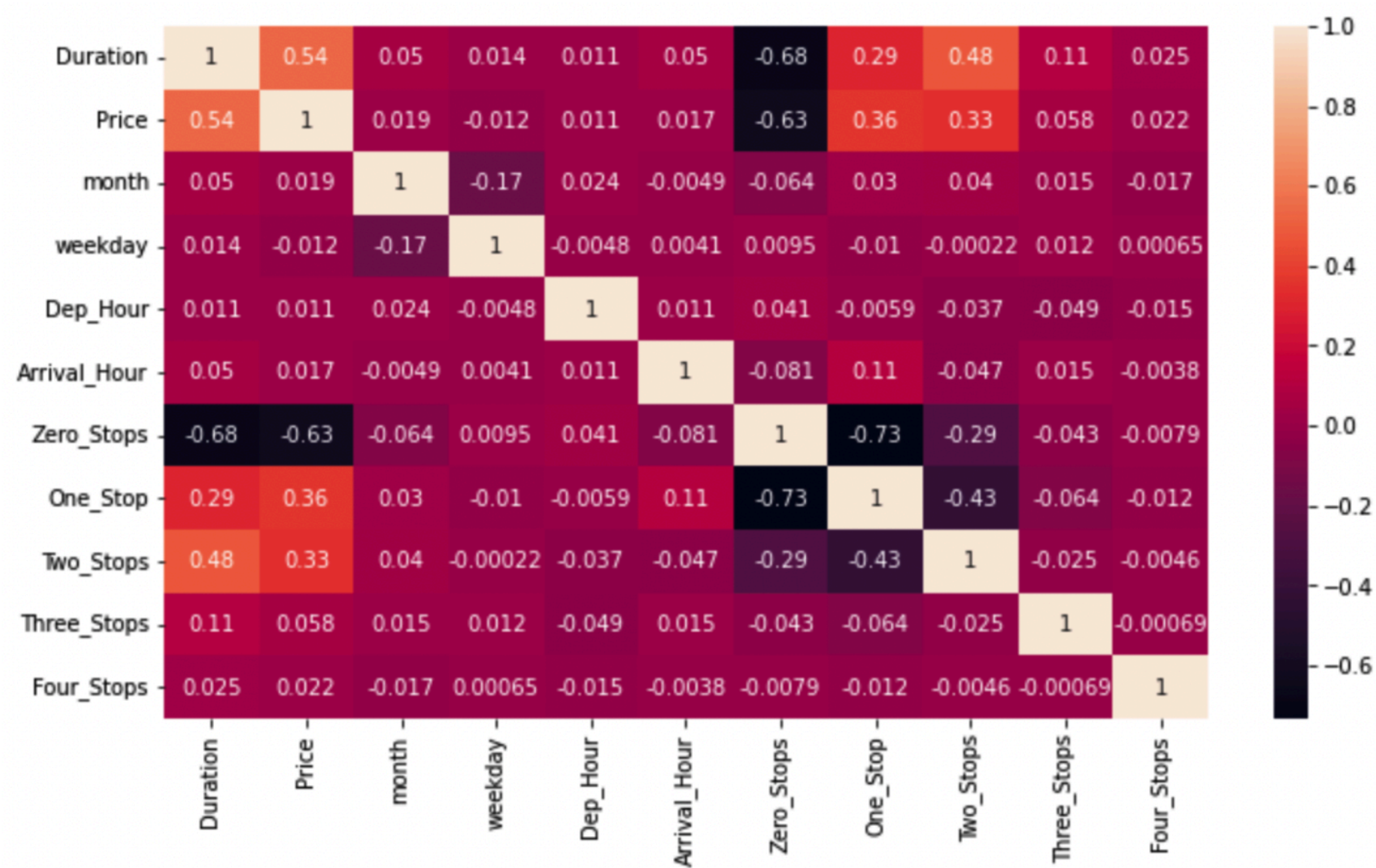
Destination	Banglore	Cochin	Delhi	Hyderabad	Kolkata	New Delhi
Source						
Banglore	NaN	NaN	4823.0	NaN	NaN	10976.5
Chennai	NaN	NaN	NaN	NaN	3850.0	NaN
Delhi	NaN	10262.0	NaN	NaN	NaN	NaN
Kolkata	9345.0	NaN	NaN	NaN	NaN	NaN
Mumbai	NaN	NaN	NaN	3342.0	NaN	NaN

pd.cut()



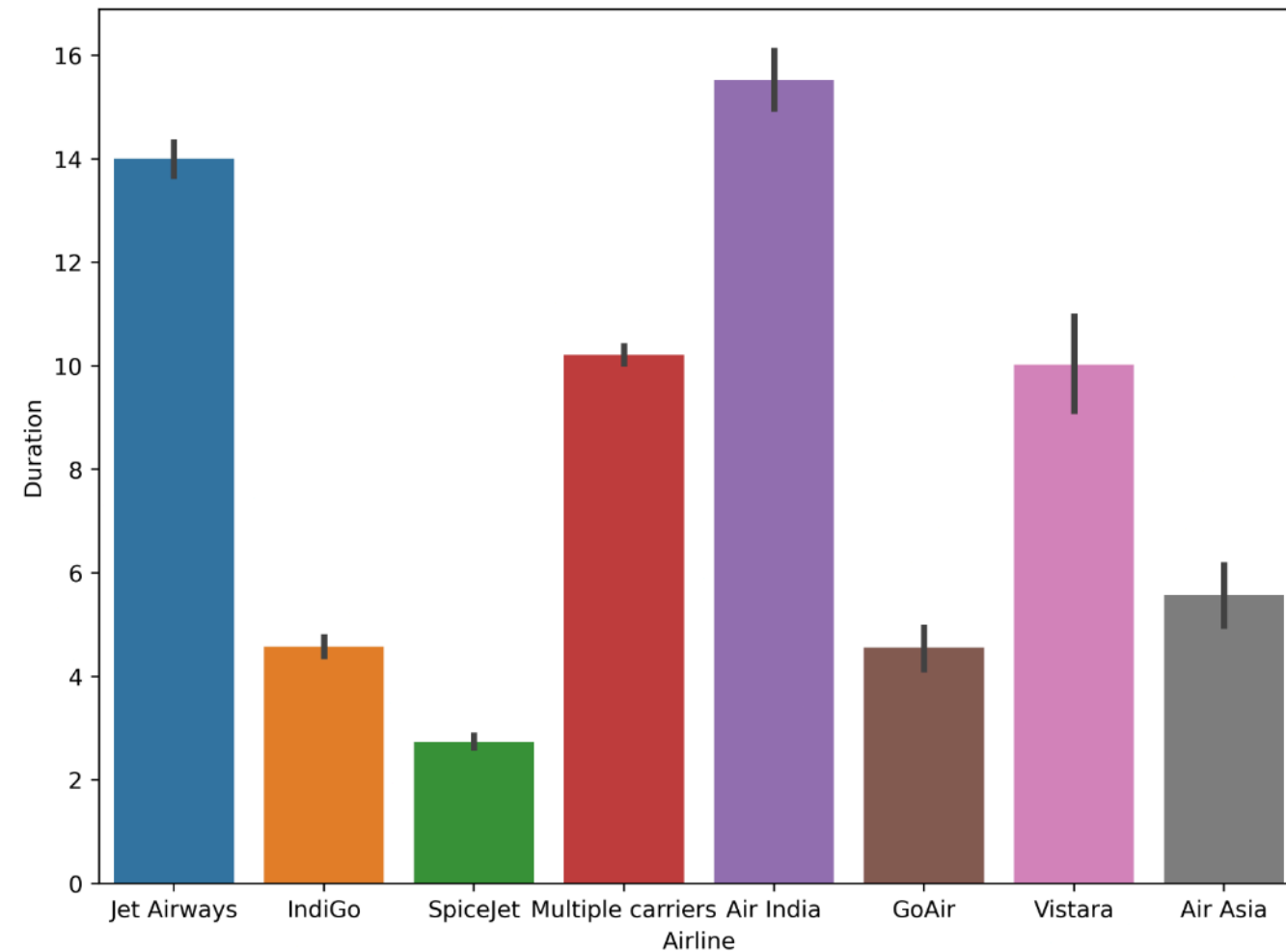
```
planes["Price_Category"] = pd.cut(planes["Price"],  
                                  labels=labels,  
                                  bins=bins)
```

Data snooping



Generating hypotheses

```
sns.barplot(data=planes, x="Airline", y="Duration")  
plt.show()
```



Next steps

- [Sampling in Python](#)
- [Hypothesis Testing in Python](#)
- [Supervised Learning with scikit-learn](#)

Congratulations!

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